

Hints and Solution: 3

Ques.1

a) $y(t) = u(t) * u(t)$

$$y(t) = \int_{-\infty}^{\infty} u(\tau)u(t - \tau)d\tau$$

$$y(t) = \int_0^t u(t - \tau)d\tau ; (t > 0)$$

$$y(t) = \int_0^t d\tau$$

$$y(t) = \begin{cases} t & \text{for } t > 0 \\ 0 & \text{for } t < 0 \end{cases}$$

$$y(t) = tu(t)$$

b) $y(t) = x(t) * \delta(t - \tau)$

$$y(t) = \int_{-\infty}^{\infty} x(m)\delta(t - m - \tau)dm$$

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau)\delta(-m(t - \tau))dm$$

$$y(t) = x(t - \tau) \int_{-\infty}^{\infty} \delta(t - m - \tau)dm$$

$$y(t) = x(t - \tau)$$

c) $y(t) = e^{\alpha}u(t) * u(t)$

$$y(t) = \int_{-\infty}^{\infty} e^{\alpha\tau}u(\tau)u(t - \tau) d\tau$$

$$y(t) = \int_0^t e^{\alpha\tau}u(\tau) d\tau$$

$$y(t) = \frac{(e^{\alpha t} - 1)}{\alpha}u(t)$$

Ques.2

$$x[n] = \delta[n] + 2\delta[n - 1] - \delta[n - 3]$$

$$h[n] = 2\delta[n + 1] + 2\delta[n - 1]$$

$$x[n] = \{1, 2, 0, -1\}$$

$$h[n] = \{2, 0, 2\}$$

a) $y_1[n] = x[n] * h[n]$

$$y_1[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k]$$

$$y_1[n] = \sum_{k=0}^3 x[k]h[n-k]$$

For

$$y_1[-2] = \sum_{k=0}^3 x[k]h[-2-k] = 0$$

$$y_1[-1] = \sum_{k=0}^3 x[k]h[-1-k] = 2$$

$$y_1[0] = \sum_{k=0}^3 x[k]h[-k] = 4$$

$$y_1[1] = \sum_{k=0}^3 x[k]h[1-k] = 2$$

$$y_1[2] = \sum_{k=0}^3 x[k]h[2-k] = 2$$

$$y_1[3] = \sum_{k=0}^3 x[k]h[3-k] = 0$$

$$y_1[4] = \sum_{k=0}^3 x[k]h[4-k] = -2$$

		$x[n]$			
	$\downarrow$	1	2	0	-1
$h[n]$	2	2	4	0	-2
	$\rightarrow 0$	0	0	0	0
	2	2	4	0	-2

$$y_1[n] = \{2, 4, 2, 2, 0, -2\}$$



b) $y_2[n] = x[n+2] * h[n]$

$$y_2[n] = x[n+2] * h[n]$$

$$y_2[n] = x[n] * \delta[n+2] * h[n]$$

$$y_2[n] = x[n] * h[n] * \delta[n+2]$$

$$y_2[n] = y_1[n] * \delta[n+2]$$

$$y_2[n] = y_1[n+2]$$

c) $y_3[n] = x[n] * h[n+2]$

$$y_3[n] = x[n] * h[n+2]$$

$$y_3[n] = x[n] * h[n] * \delta[n+2]$$

$$y_3[n] = y_1[n] * \delta[n+2]$$

$$y_3[n] = y_1[n+2]$$

Ques.3

a) False To prove

$$x[n] * \{h[n]g[n]\} = \{x[n] * h[n]\}g[n]$$

$$\begin{aligned} \text{LHS} &= x[n] * \{h[n]g[n]\} \\ &= \sum x[k]h[n-k]g[n-k] \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \{x[n] * h[n]\}g[n] \\ &= \left\{ \sum x[k]h[n-k] \right\} g[n] \end{aligned}$$

LHS $\neq$ RHS

Counter example assume $g[n] = \delta[n]$

b) True.

$$x(at) * h(at) = \int_{-\infty}^{\infty} x(a\tau)h(at - a\tau)d\tau$$

Let $m = a\tau$. Perform change of variables.

For $a > 0$ (eg. $a=2$):

$$\begin{aligned} x(at) * h(at) &= \frac{1}{a} \int_{-\infty}^{\infty} x(m)h(at - m)dm \\ &= \frac{1}{a} y(at) \\ y(at) &= ax(at) * h(at) \end{aligned}$$

In general:

$$y(at) = |a|x(at) * h(at)$$

c) True

$$y(t) = x(t) * h(t)$$

$$y(-t) = x(-t) * h(-t)$$

$$y(-t) = \{-x(t)\} * \{-h(t)\}; \text{ Since they are odd signals}$$

$$y(-t) = x(t) * h(t)$$

$$y(-t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau = y(t)$$

Hence $y(t)$ is even signal.

d) $y(t) = x(t) * h(t)$

$$E_V\{y(t)\} = x(t) * E_V\{h(t)\} + E_V\{x(t)\} * h(t)$$

If this is true,

$$\begin{aligned} x(t) * \left[\frac{h(t)}{2} \right] + \frac{[x(-t) * h(-t)]}{2} \\ = \frac{[x(t) * h(t)]}{2} + x(t) * \left[\frac{h(-t)}{2} \right] + x(t) * \left[\frac{h(t)}{2} \right] + \frac{[x(-t) * h(t)]}{2} \end{aligned}$$

$$\text{Thus } x(-t) * h(-t) = x(t) * h(-t) + x(t) * h(t) + x(-t) * h(t)$$

Thus, in general, the given property is false.

Counter example

$$x(t) = \delta(t - 1)$$

$$h(t) = \delta(t + 1)$$

$$y(t) = \delta(t)$$

Ques.4

$\widetilde{x}_1(t)$ & $\widetilde{x}_2(t)$ is periodic $T = T_0$

a) $\widetilde{y}(t) = \widetilde{x}_1(t) * \widetilde{x}_2(t)$

$$\widetilde{y}(t) = \int_0^{T_0} \widetilde{x}_1(\tau) \widetilde{x}_2(t - \tau) d\tau$$

For proving $\widetilde{y}(t)$ is periodic with $T = T_0$

$\widetilde{y}(t) = \widetilde{y}(t + T_0)$ must satisfy

$$\widetilde{y}(t + T_0) = \int_0^{T_0} \widetilde{x}_1(\tau) \widetilde{x}_2(t + T_0 - \tau) d\tau = \widetilde{y}(t)$$

Hence proved.

b) $\widetilde{y}_a(t) = \int_a^{a+T_0} \widetilde{x}_1(\tau) \widetilde{x}_2(t - \tau) d\tau$

Put $a = kT_0 + b$ $0 \leq b \leq T_0$

$$= \int_{kT_0+b}^{kT_0+b+T_0} \widetilde{x}_1(\tau) \widetilde{x}_2(t - \tau) d\tau$$

k is integer and so taking $k = 1, 2, 3 \dots$ also does not change the equation

$$\widetilde{y}_a(t) = \int_b^{b+T_0} \widetilde{x}_1(\tau) \widetilde{x}_2(t - \tau) d\tau$$

$$\widetilde{y}_a(t) = \int_b^{T_0} \widetilde{x}_1(\tau) \widetilde{x}_2(t - \tau) d\tau + \int_{T_0}^{b+T_0} \widetilde{x}_1(\tau) \widetilde{x}_2(t - \tau) d\tau$$

Substitute $\tau = \tau - T_0$ in second term

$$\widetilde{y}_a(t) = \int_b^{T_0} \widetilde{x}_1(\tau) \widetilde{x}_2(t - \tau) d\tau + \int_0^b \widetilde{x}_1(\tau) \widetilde{x}_2(t - \tau) d\tau$$

$$\widetilde{y}_a(t) = \int_0^{T_0} \widetilde{x}_1(\tau) \widetilde{x}_2(t - \tau) d\tau$$

$$\widetilde{y}_a(t) = \widetilde{y}(t)$$

Hence proved.

Ques.5

a) $\varphi[n] = \left(\frac{1}{2}\right)^n u[n]$

$$\frac{1}{2} \varphi[n - 1] = \left(\frac{1}{2}\right)^n u[n - 1]$$

$$\varphi[n] - \frac{1}{2} \varphi[n - 1] = \left(\frac{1}{2}\right)^n \{u[n] - u[n - 1]\}$$

$$\left(\frac{1}{2}\right)^n \{u[n] - u[n - 1]\} = \begin{cases} 0 & \text{for } n \neq 0 \\ 1 & \text{for } n = 0 \end{cases}$$

$$\left(\frac{1}{2}\right)^n \{u[n] - u[n - 1]\} = \delta[n]$$

So
$$\delta[n] = \varphi[n] - \left(\frac{1}{2}\right) \varphi[n-1]$$

In general

$$\begin{aligned} x[n] &= \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \\ &= \sum_{k=-\infty}^{\infty} x[k] \left\{ \varphi[n-k] - \frac{1}{2} \varphi[n-k-1] \right\} \\ &= \sum_{k=-\infty}^{\infty} \left(x[k] - \frac{1}{2} x[k-1] \right) \varphi[n-k] \\ &= a_k = x[k] - \frac{1}{2} x[k-1] \\ &= \sum_{k=-\infty}^{\infty} a_k \varphi[n-k] \end{aligned}$$

b)

$$y[n] = \sum_{k=-\infty}^{\infty} a_k r[n-k]$$

c) $y[n] = x[n] * h[n]$

$$\begin{aligned} \delta[n] &= \varphi[n] - \frac{1}{2} \varphi[n-1] \\ h[n] &= r[n] - \frac{1}{2} r[n-1] \\ &= \Psi(n) * r[n] \end{aligned}$$

where

$$\begin{aligned} \Psi(n) &= \delta[n] - \frac{1}{2} \delta[n-1] \\ y[n] &= x[n] * \Psi[n] * r[n] \end{aligned}$$

d) Already shown that

$$\begin{aligned} h[n] &= r[n] - \frac{1}{2} r[n-1] \\ \Psi(n) * \varphi(n) &= \left[\delta(n) - \frac{1}{2} \delta(n-1) \right] * \varphi(n) = \varphi(n) - \frac{1}{2} \varphi(n-1) = \delta(n) \end{aligned}$$

Ques.6

a) Received signal is delayed and attenuated.

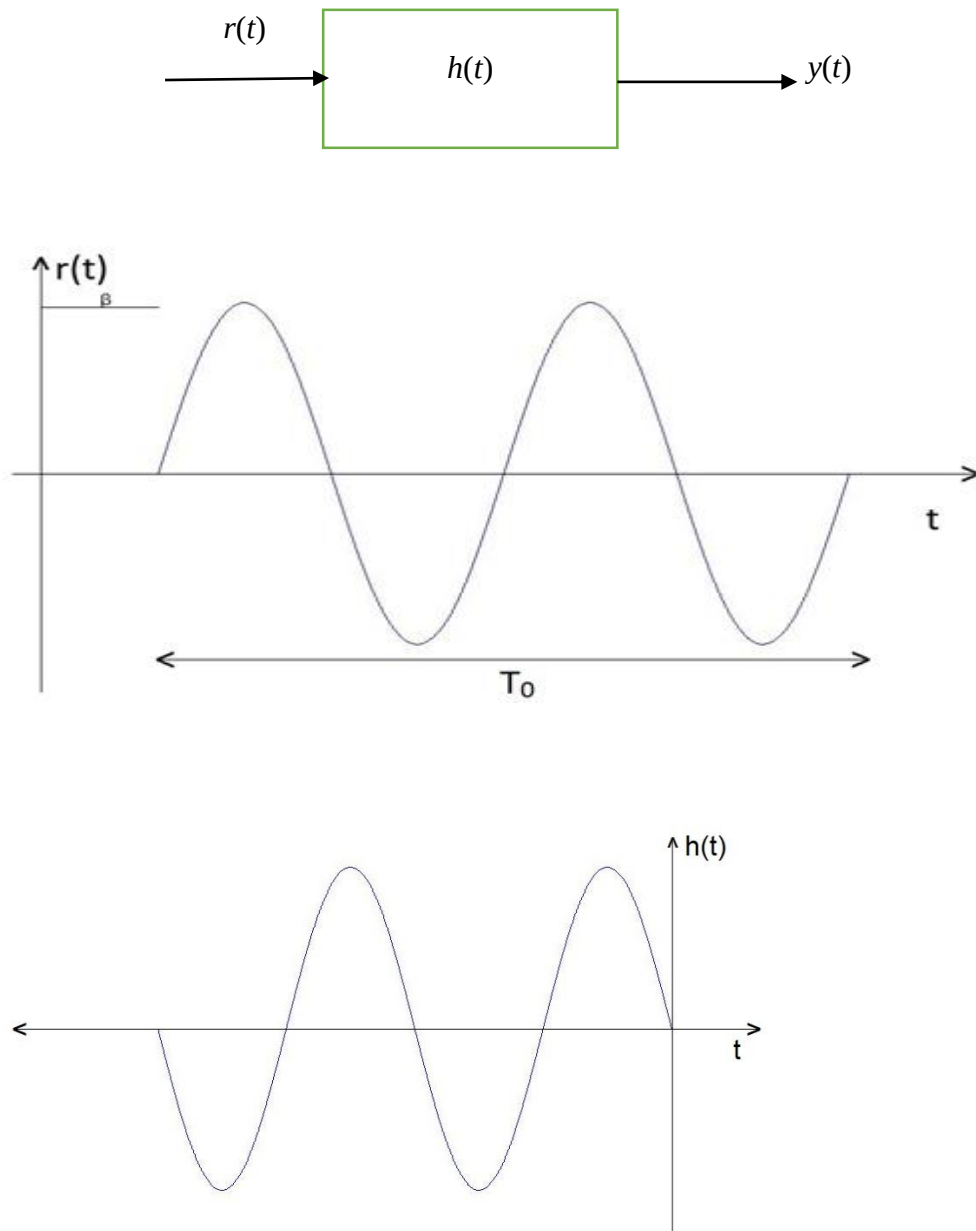
So, $r(t) = \alpha \sin(\omega_c t - \beta)$

where α is attenuating factor and β is delay

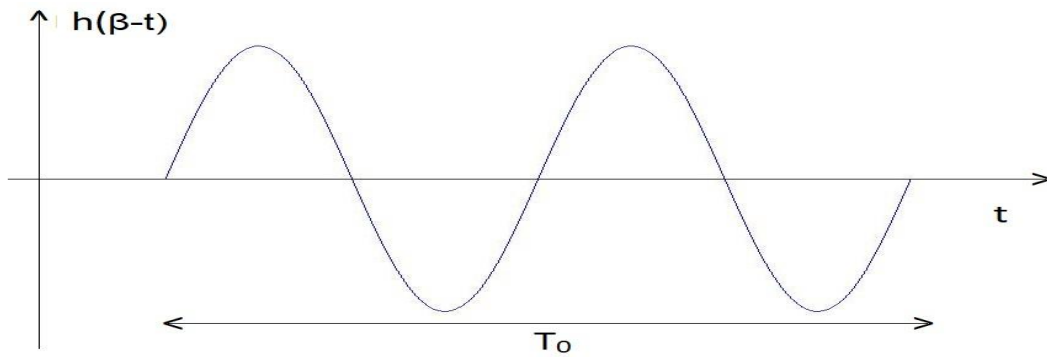
$$\begin{aligned} x(t) * h(t) &= x(t) * \alpha \delta(t - \beta) \\ &= \alpha x(t - \beta) \\ &= \alpha \sin(\omega_c t - \beta) \end{aligned}$$

Hence $h(t) = \alpha \delta(t - \beta)$ is the correct impulse response.

- b) Matched filter is commonly used for detecting signals in the presence of noise. Matched filter response is the time-reversed version of the transmitted signal. Let us investigate what happens when the reflected signal passes through a matched filter.

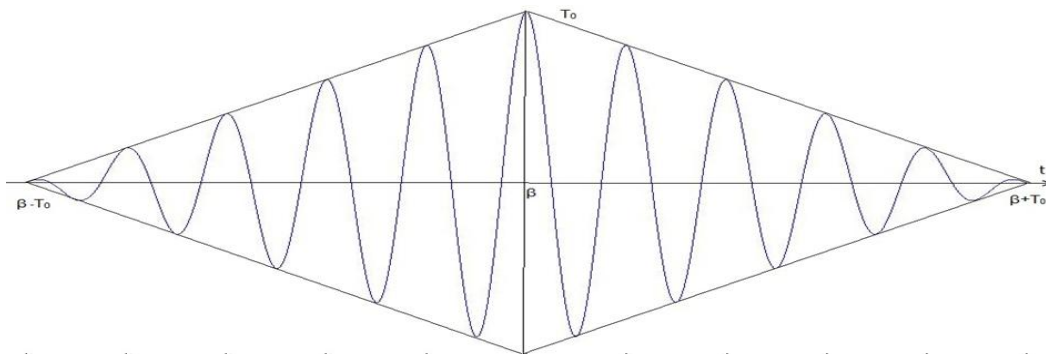


For convolution, we take the reversed version of the impulse response and shift it through various t .



It is intuitively clear that the output will be maximum when $h(t)$ is reversed and shifted by β , because it leads to maximum overlap. Thus, by detecting the time at which output becomes maximum, we calculate β .

$y(t)$ can also be computed analytically.



Ques.7

a)

$$w[n] = \frac{1}{2}w[n-1] + x[n]$$

$$y[n] = \alpha y[n-1] + \beta w[n]$$

$$w[n] = \frac{1}{\beta} \{y[n] - \alpha y[n-1]\} \dots \dots (1)$$

$$\frac{1}{2}w[n-1] = \frac{1}{2\beta} \{y[n-1] - \alpha y[n-2]\} \dots \dots (2)$$

From (1) and (2)

$$w[n] - \frac{1}{2}w[n-1] = \frac{y[n]}{\beta} - \left(\alpha + \frac{1}{2}\right) \frac{y[n-1]}{\beta} + \frac{\alpha y[n-2]}{2\beta}$$

$$y[n] = \beta x[n] - \frac{\alpha}{2} y[n-2] + \left(\alpha + \frac{1}{2}\right) y[n-1]$$

Thus $\beta = 1$ and $\alpha = \frac{1}{4}$

b)

$$w[n] = \frac{1}{2} w[n-1] + x[n]$$

Evaluate the impulse response

$$w[0] = \frac{1}{2} w[-1] + \delta[0]$$

$w[-1] = 0$ {System is causal LTI}

$$w[0] = \delta[0] = 1$$

$$w[1] = \frac{1}{2} w[0] = \frac{1}{2}$$

$$w[2] = \frac{1}{4} w[0] = \frac{1}{4}$$

$\vdots$

Thus

$$w[n] = \left(\frac{1}{2}\right)^n u[n]$$

$$h_1[n] = \left(\frac{1}{2}\right)^n u[n]$$

Similarly

$$h_2[n] = \left(\frac{1}{4}\right)^n u[n]$$

$$h[n] = h_1[n] * h_2[n]$$

$$= \sum_{k=-\infty}^{k=\infty} \left(\frac{1}{2}\right)^k u[k] \left(\frac{1}{4}\right)^{n-k} u[n-k]$$

$$= \sum_{k=0}^{k=n} \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k} ; \text{for } n > 0$$

$$h[n] = \left(\frac{1}{2}\right)^{2n} [2^{n+1} - 1] u[n]$$

Ques.8

a) True

$$\int_{-\infty}^{\infty} |h(t)| dt = \sum_{k=-\infty}^{k=\infty} \int_{KT}^{KT+T} |h(t)| dt = \infty$$

b) False

$$h(t) = \delta(t - 1)$$

$$h^{-1}(t) = \delta(t + 1)$$

$h^{-1}(t)$ is non causal system.

c) False. Suppose $h[n] = u[n]$ then

$$\sum_{k=-\infty}^{\infty} |h[k]| = \sum_{k=-\infty}^{\infty} 1 = \infty$$

d) False if $h[n] = \infty$ then the system is unstable.

e) False. $h(t) = u(t)$ then

$$\int_{-\infty}^{\infty} |h(t)| dt = \int_0^{\infty} dt = \infty$$

f) False

$$h_1(t) = \delta(t + 1) \quad [non - causal]$$

$$h_2(t) = \delta(t - 1) \quad [causal]$$

$$h(t) = h_1(t) * h_2(t) = \delta(t) \quad [causal]$$

which is causal.

g) False

For example, $h(t) = e^{-t}u(t)$

$$s(t) = \int_{-\infty}^t e^{-\tau} u(\tau) d\tau$$

$$= (1 - e^{-t})u(t)$$

$$\int_{-\infty}^{\infty} |s(\tau)| d\tau = \int_0^{\infty} (1 - e^{-\tau}) d\tau = \infty$$

h) Of course, a causal system only respond after an input is applied. Thus,
 $s[n] = 0$ For $n < 0$

Ques.9

$$h(t) = \{ \{ h_1 * h_2 + h_2 * h_2 - h_2 * h_1 \} * h_1 + h_1^{-1} \} * h_2^{-1}$$

$$h(t) = \{ h_1 + h_2 - h_1 \} * h_1 + h_1^{-1} * h_2^{-1}$$

$$h(t) = h_2 * h_1 + h_1^{-1} * h_2^{-1}$$

Ques.10

$$(D^2 + 3D + 2)y(t) = Dx(t) = u(t)$$

Let us solve

$$(D^2 + 3D + 2)y(t) = Dx(t) = u(t)$$

The output $y(t) = y_p(t) + y_n(t)$

$$y_n(t) = Ae^{st}$$

$$S^2 + 3S + 2 = 0$$

$$S = -1, S = -2$$

$$y_n(t) = C_1 e^{-t} + C_2 e^{-2t}$$

$$y_p(t) = B$$

$$2B = 1$$

$$B = 1/2$$

Assuming system is causal LTI

$$y(t) = \left(\frac{1}{2} + C_1 e^{-t} + C_2 e^{-2t} \right) u(t)$$

$$y(0^+) = 0 \dots \dots (1)$$

$$\frac{dy(0^+)}{dt} = 0 \dots \dots (2)$$

From 1,

$$\frac{1}{2} + C_1 + C_2 = 0$$

From 2,

$$-C_1 - 2C_2 = 0$$

$$C_1 = -1, C_2 = 1/2$$

$$y(t) = \left(\frac{1}{2} - e^{-t} + 1/2 e^{-2t} \right) u(t)$$

$$h(t) = \frac{d^2 y(t)}{dt^2} = \frac{d}{dt} \{e^{-t} - e^{-2t}\}$$

$$h(t) = (-e^{-t} + e^{-2t})u(t)$$